

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12409546
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Koo; Donghan et al.

Mobile robot apparatus

Abstract

A mobile robot apparatus is provided. The mobile robot apparatus includes: a body; a first wheel disposed at a first side surface of the body; a second wheel disposed at a second side surface of the body opposite to the first side surface; a first drive device configured to provide a driving force to each of the first wheel and the second wheel; a second drive device configured to move the body in a vertical direction relative to at least one of a first center axis of the first wheel and a second center axis of the second wheel; and a processor configured to control the second drive device to move the body to contact a surface on which the mobile robot apparatus is disposed, or move the body away from the surface on which the mobile robot apparatus is disposed.

Inventors: Koo; Donghan (Suwon-si, KR), Hyun; Sangmin (Suwon-si, KR), Kim; Yongtae (Suwon-si, KR), Kuk; Junggap (Suwon-si, KR), Choi; Kuyoung (Suwon-si, KR)

Applicant: SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)

Family ID: 79554418

Assignee: SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)

Appl. No.: 18/077003

Filed: December 07, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230098075 A1	Mar. 30, 2023

Foreign Application Priority Data

KR	10-2020-0087366	Jul. 15, 2020
KR	10-2020-0144675	Nov. 02, 2020

Related U.S. Application Data

continuation parent-doc WO PCT/KR2021/008303 20210630 PENDING child-doc US 18077003

Publication Classification

Int. Cl.: **B25J5/00** (20060101); **B25J9/10** (20060101); **B25J9/12** (20060101); **B25J9/16** (20060101); **B25J13/08** (20060101); **B25J19/02** (20060101)

U.S. Cl.:

CPC **B25J5/007** (20130101); **B25J9/1035** (20130101); **B25J9/126** (20130101); **B25J9/1664** (20130101); **B25J13/089** (20130101); **B25J19/021** (20130101);

Field of Classification Search

CPC: B25J (5/007); B25J (9/126); B25J (9/1035); B25J (9/102); B25J (19/023); B25J (5/00)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7468592	12/2007	Lim et al.	N/A	N/A
9532690	12/2016	Jang et al.	N/A	N/A
10168715	12/2018	Tsuji et al.	N/A	N/A
10307911	12/2018	Higuchi	N/A	B25J 9/1697
10507400	12/2018	Higuchi	N/A	N/A
10799806	12/2019	Sasagawa	N/A	G01P 15/18
2004/0182614	12/2003	Wakui	N/A	N/A
2007/0132420	12/2006	Lim et al.	N/A	N/A
2007/0215494	12/2006	Yuhara	206/37	A45D 33/006
2008/0084175	12/2007	Hollis	901/1	G05D 1/0891
2015/0105940	12/2014	Takase	N/A	N/A
2015/0208891	12/2014	Jang et al.	N/A	N/A
2019/0015758	12/2018	Higuchi	N/A	N/A
2019/0384325	12/2018	Lee et al.	N/A	N/A
2020/0070923	12/2019	Roufas	N/A	N/A
2020/0180141	12/2019	Dong	N/A	N/A
2020/0262053	12/2019	Clerc	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
61-11082	12/1985	JP	N/A
2008-272859	12/2007	JP	N/A
2010-264555	12/2009	JP	N/A
2019-18276	12/2018	JP	N/A
2001-0089835	12/2000	KR	N/A
10-0783134	12/2006	KR	N/A
10-2011-0108103	12/2010	KR	N/A

10-2011-0132745	12/2010	KR	N/A
10-2018-0083398	12/2017	KR	N/A
10-1912806	12/2017	KR	N/A
10-1984012	12/2018	KR	N/A
10-2020-0025102	12/2019	KR	N/A
10-2125334	12/2019	KR	N/A

OTHER PUBLICATIONS

Communication issued on Oct. 27, 2023 by European Patent Office in European Patent Application No. 21842616.1. cited by applicant

S. Kim and S. Kwon, "Nonlinear Optimal Control Design for Underactuated Two-Wheeled Inverted Pendulum Mobile Platform", IEEE/ASME Transactions on Mechatronics, vol. 22, No. 6, Dec. 2017, pp. 2803-2808, (6 pages total). cited by applicant

Written Opinion (PCT/ISA/237) issued Oct. 18, 2022 by the International Searching Authority in International Application No. PCT/KR2021/008303. cited by applicant

International Search Report (PCT/ISA/210) issued Oct. 18, 2021 by the International Searching Authority in International Application No. PCT/KR2021/008303. cited by applicant

Communication dated Apr. 2, 2025, issued by Korean Intellectual Property Office in Korean Patent Application No. 10-2020-0144675. cited by applicant

Primary Examiner: Tran; Dalena

Attorney, Agent or Firm: Sughrue Mion, PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a by-pass continuation application of International Application No. PCT/KR2021/008303, filed on Jun. 30, 2021, which is based on and claims priority to Korean Patent Application No. 10-2020-0087366, filed on Jul. 15, 2020, and Korean Patent Application No. 10-2020-014675, filed on Nov. 2, 2020, in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entireties.

BACKGROUND

1. Field

(1) The disclosure relates to a mobile robot apparatus, and more particularly, to a mobile robot apparatus which can travel stably and collect information as a body travels contacting with a surface, or travels spaced apart from the surface.

2. Description of Related Art

(2) With developments in robot technology, robots are not only found in specialized academic fields or industrial fields which require mass scale labor, but they are also becoming more common in average homes. In addition, robots are not only being supplied to perform a function at a fixed position, but mobile robots are also being supplied that are capable of traveling. Among the mobile robots, two wheeled robots with two wheels have an advantage of taking up a small surface area, and two wheeled robots of various types (Segway, Ninebot, etc.) are widely utilized.

(3) In the related art, two wheeled robots are configured such that a center of mass of the body is always above a rotational axis of a wheel relative to the surface on which the robot may travel upon, and a balancing control is carried out so as to prevent the robot from falling. The mobile robots embodying this structure have a problem in that they are susceptible to swaying according to

a rotational inertial force which can make it difficult to stably travel, and this instability further prevents such robots from being able to return to their original position when having fallen.

SUMMARY

- (4) Provided is a mobile robot apparatus which can travel stably and collect information as a body travels contacting with a surface, or travels spaced apart from the surface
- (5) Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.
- (6) According to an aspect of the disclosure, a mobile robot apparatus may include: a body; a first wheel disposed at a first side surface of the body; a second wheel disposed at a second side surface of the body opposite to the first side surface; a first drive device configured to provide a driving force to each of the first wheel and the second wheel; a second drive device configured to move the body in a vertical direction relative to at least one of a first center axis of the first wheel and a second center axis of the second wheel; and a processor configured to control the second drive device to move the body to contact a surface on which the mobile robot apparatus is disposed, or move the body away from the surface on which the mobile robot apparatus is disposed.
- (7) The first drive device may include: a first motor configured to provide a first driving force to the first wheel, and a second motor configured to provide a second driving force to the second wheel.
- (8) The second drive device may include a rack provided in at least one of the first wheel and the second wheel, and a pinion provided in the body and engaged with the rack.
- (9) The rack may be arranged vertically relative to the surface on which the mobile robot apparatus is disposed.
- (10) The rack may include: a first rack provided in the first wheel; and a second rack provided in the second wheel, and the pinion may include: a first pinion engaged with the first rack, and a second pinion engaged with the second rack.
- (11) The second drive device further may include: a third motor configured to rotate the first pinion, and a fourth motor configured to rotate the second pinion.
- (12) The first wheel may include: a rotatable first wheel cover; and a first intermediate member provided between the first wheel cover and the body, the second wheel may include: a rotatable second wheel cover; and a second intermediate member provided between the second wheel cover and the body, the first rack may be provided on the first intermediate member, and the second rack may be provided on the second intermediate member.
- (13) The mobile robot apparatus may further include a guide rail provided in at least one of the first intermediate member and the second intermediate member, and the body further may include a block configured to be movable along the guide rail.
- (14) The guide rail may be oriented vertically relative to the surface on which the mobile robot apparatus is disposed.
- (15) The mobile robot apparatus may further include a camera provided at the body and configured to capture an image of a surrounding environment of the mobile robot apparatus.
- (16) The mobile robot apparatus may further include a caster provided at a bottom surface of the body and configured to be rotatable.
- (17) A rotational axis of the caster may be parallel with the first center axis of the first wheel or the second center axis the second wheel.
- (18) The processor may be further configured to control the second drive device to selectively cause the caster to come into contact with or to be spaced apart from the surface on which the mobile robot apparatus is disposed.
- (19) The processor may be further configured to control the second drive device to tilt the body in a direction to which the mobile robot apparatus is turning.
- (20) The processor may be further configured to control the second drive device, based on the body contacting the surface on which the mobile robot apparatus is disposed, to move the first and

second wheels toward the body until a center of mass of the body is positioned outside a section between a contact point of the body with the surface on which the mobile robot apparatus is disposed and a contact point of the first and second wheels with the surface on which the mobile robot apparatus is disposed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other aspects, features, and advantages of certain embodiments of the present disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is a front perspective view of a mobile robot apparatus according to an embodiment of the disclosure;

(3) FIG. 2 is a rear perspective view of a mobile robot apparatus according to an embodiment of the disclosure;

(4) FIG. 3 is a front view of a mobile robot apparatus of which a body is in contact with a surface;

(5) FIG. 4 is a front perspective view of a mobile robot apparatus of which a body travels to an upper side;

(6) FIG. 5 is a front view of a mobile robot apparatus of which a body is spaced apart from a surface;

(7) FIG. 6 is an exploded, front, perspective view of a mobile robot apparatus according to an embodiment of the disclosure;

(8) FIG. 7 is a cross-sectional view of the mobile robot apparatus shown in FIG. 1 taken along line I-I;

(9) FIG. 8 is a side, interior view of a first wheel according to an embodiment of the disclosure;

(10) FIG. 9 is a block view schematically illustrating a control process of a mobile robot apparatus according to an embodiment of the disclosure;

(11) FIG. 10 is a cross-sectional view of the mobile robot apparatus shown in FIG. 1 taken along line II-II;

(12) FIG. 11 is a cross-sectional view of the mobile robot apparatus shown in FIG. 4 taken along line III-III;

(13) FIG. 12 and FIG. 13 are views illustrating a body being tilted when a mobile robot apparatus is turned; and

(14) FIG. 14 to FIG. 17 are views illustrating a process of a body returning to its original position when having fallen.

DETAILED DESCRIPTION

(15) Embodiments described below are provided as examples to assist in the understanding of the disclosure, and it is to be understood that the disclosure may be variously modified and realized, differently from the embodiments described herein. In addition, accompanied drawings are not shown in its actual scale but shown with scales of some elements exaggerated to assist in the understanding of the disclosure.

(16) Terms used in the disclosure and in the claims are general terms selected considering their function herein. However, the terms may change depending on intention, legal or technical interpretation, emergence of new technologies, and the like of those skilled in the related art. Further, in certain cases, there may be terms arbitrarily selected. With respect to these terms, the meaning of the term may be interpreted as defined in the description, and if there is no specific definition of the term, the term may be interpreted based on the overall context of the disclosure and the technical common sense of the related art.

(17) In the disclosure, expressions such as “have,” “may have,” “include,” “may include,” or the

like are used to designate a presence of a corresponding characteristic (e.g., elements such as numerical value, function, operation, or component), and not to preclude a presence or a possibility of additional characteristics. As used herein, expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list. For example, the expression, “at least one of a, b, and c,” should be understood as including only a, only b, only c, both a and b, both a and c, both b and c, or all of a, b, and c.

(18) Further, elements necessary in describing each embodiment of the disclosure are described herein, but the disclosure is not necessarily limited thereto. Accordingly, some elements may be modified or omitted, and other elements may be added. In addition, the elements may be distributed and disposed in separate devices different from one another.

(19) Furthermore, embodiments of the disclosure have been described in detail with reference to the accompanied drawings and descriptions described in the accompanied drawings below, it is to be understood that the disclosure is not limited by or limited to the embodiments.

(20) Embodiments will be described in detail below with reference to the accompanied drawings.

(21) FIG. 1 is a front perspective view of a mobile robot apparatus according to an embodiment of the disclosure. FIG. 2 is a rear perspective view of the mobile robot apparatus according to an embodiment of the disclosure. FIG. 3 is a front view of a mobile robot apparatus of which a body is in contact with a surface.

(22) A mobile robot apparatus 1 may be a device with various functions such as autonomously traveling while recognizing a surrounding environment and collecting information, delivering information to a user, and the like. The mobile robot apparatus 1 may recognize the surrounding environment based on a voice, a sound, and an image. In addition, information may be delivered to the user by controlling other electronic products through wireless communication or outputting a voice.

(23) The mobile robot apparatus 1 may collect and analyze various information such as sounds, voices, images, and the like from the surrounding environment to make stable autonomous traveling possible. For example, the mobile robot apparatus 1 may include a microphone, a camera, a sensor, and the like to collect information on the surrounding environment.

(24) The mobile robot apparatus 1 may physically travel by including a drive member, and through the drive member, execute various functions of the mobile robot apparatus 1 throughout the environment of the user which may include indoors and outdoors.

(25) If the mobile robot apparatus 1 is used within a home, the mobile robot apparatus may interact with the electronic products within the home, such as a television (TV), a cleaner, and a washer, to execute a function and collect information, and to deliver the collected information to family members including a pet. Accordingly, the mobile robot apparatus may connect all members within the home with the electronic products.

(26) The mobile robot apparatus 1 may continuously check and inspect the environment within the home even when the user is not present and connect the user with family members including a pet that needs assistance. In addition, other electronic products within the home may be checked and operated through physical travel of the mobile robot apparatus 1. Through the above, safety within the home may be sought after and security within the home may be enhanced.

(27) The mobile robot apparatus 1 according to an embodiment of the disclosure may be implemented in a form related to the performance of a job within the home, but is not limited thereto, and may be implemented as a robot apparatus according to various embodiments.

(28) Referring to FIGS. 1 to 3, the mobile robot apparatus 1 according to an embodiment of the disclosure may include a body 100, a first wheel 200, and a second wheel 300. Covers of the body 100, the first wheel 200, and the second wheel 300 may have a same curvature, and accordingly, an exterior of the mobile robot apparatus 1 may be formed to realize a sphere.

(29) Although not shown in FIG. 1 to FIG. 3, a motor, a battery, an actuator, a gear, a bearing, and the like for driving the mobile robot apparatus 1 may be included inside of the body 100 of the

mobile robot apparatus **1**.

(30) A camera **101** may be provided at an outer surface **105** of the body **100** to capture the surrounding environment of the mobile robot apparatus **1**. In addition, a sensor **102** may be provided at the outer surface **105** of the body **100**. The sensor **102** may be at least one from among an image sensor which detects an obstacle, a sound sensor which detects a voice, a temperature sensor which detects temperature, and a humidity sensor which detects humidity.

(31) The mobile robot apparatus **1** may recognize, based on information collected from the camera **101** and from at least one sensor **102**, the surrounding environment, autonomously travel and collect information, and deliver information to the user.

(32) A caster **103** may be rotatably connected to a bottom surface **107** of body **100**. A rotational axis **X2** of the caster **103** may be parallel with a center axis **X1** of the first and second wheels **200** and **300**.

(33) The caster **103** may be configured such that a portion of caster **103** is protruded to the outside of the bottom surface of the body **100**. The caster **103** may be formed of plastic or metal, may have a rotatable ball caster form and thereby, may reduce frictional drag with the surface **150** upon which the mobile robot apparatus **1** may travel.

(34) Each of the first and second wheels **200** and **300** may be in contact with the surface **150** from a first point **S1** and a second point **S2**. In addition, the bottom surface **107** of the body **100**, excluding the caster **103**, may be spaced apart from the surface **150**, and the caster **103** may be in contact with the surface **150** from a third point **S3**.

(35) Accordingly, the mobile robot apparatus **1** may be supported from the first to third points **S1**, **S2**, and **S3** of the surface **150** (three-point support mode). The body **100** may be in contact with the surface **150**, and the center of mass of the body **100** may be positioned below the center axis **X1** of the first and second wheels **200** and **300**.

(36) When the mobile robot apparatus **1** completes traveling to a specific location, the center of mass of the body **100** may travel to a point below the center axis **X1** of the first and second wheels **200** and **300**, and the mobile robot apparatus **1** may stop at the specific location in a stable position.

(37) When the center of mass of the body **100** is at a point below the center axis **X1** of the first and second wheels **200** and **300**, the mobile robot apparatus **1** may travel stably at a low speed without swaying. In addition, the camera **101** disposed on the unswaying body **100** may accurately capture the surrounding environment of the mobile robot apparatus **1**.

(38) In addition, because there is no need for control of a second drive device **500** (see FIG. 9) that moves the body **100**, which will be described below, in a vertical direction based on the center axis of the first and second wheels **200** and **300**, the mobile robot apparatus **1** may efficiently travel at low energy.

(39) The first wheel **200** and the second wheel **300** may be respectively disposed at opposite side surfaces of the body **100**. The first and second wheels **200** and **300** may respectively be provided with driving force from a first drive device **400** (FIG. 6) and rotate about the center axis **X1**. The center axis **X1** of the first and second wheels **200** and **300** may be horizontal to the surface **150**.

(40) The mobile robot apparatus **1** may travel toward a front direction or a rear direction according to the first and second wheels **200** and **300** rotating about the **X1** axis, or turn to travel freely to a desired location.

(41) FIG. 4 is a front perspective view of the mobile robot apparatus where the body **100** is raised upward and away from the surface **150**. FIG. 5 is a front view of the mobile robot apparatus **1** where the body **100** is raised upward and away from the surface **150**.

(42) Referring to FIG. 4 and FIG. 5, the body **100** may move in a vertical direction upward relative to the center axis **X1** of the first and second wheels **200** and **300**. Accordingly, a gap distance with the surface **150** may be further increased for the body **100**, and the center of mass of the body **100** may be positioned above the center axis **X1** of the first and second wheels **200** and **300**.

(43) That is, because both the bottom surface **107** of the body **100** and the caster **103** are not in

contact with the surface **150**, and only the first and second wheels **200** and **300** are in contact with the surface **150** from the first and second points **S1** and **S2**, the mobile robot apparatus **1** may travel using the first and second points **S1** and **S2** of the surface **150** in a supported state (two-point support mode).

(44) Accordingly, the body **100** may travel efficiently without the body **100** contacting the surface **150** even on surfaces with cushioning such as a carpet because sufficient clearance is secured from the surface **150**. In addition, because sufficient distance to an extent of passing an obstacle between the body **100** and the surface **150** is secured, the mobile robot apparatus **1** may easily travel while avoiding collisions between the body **100** and the obstacles.

(45) FIG. **6** is an exploded perspective view of a mobile robot apparatus **1** according to an embodiment of the disclosure. FIG. **7** is a top, cross-sectional view of the mobile robot apparatus **1** shown in FIG. **1** taken along line I-I. FIG. **8** is a side view of a first wheel (viewing the interior side of the wheel which faces body **100** when the wheel is attached) according to an embodiment of the disclosure. FIG. **9** is a block view schematically illustrating a control process of the mobile robot apparatus according to an embodiment of the disclosure.

(46) Referring to FIGS. **6** to **9**, the first wheel **200** may include a first wheel cover **210** and a first intermediate member **220**, and a second wheel **300** may include a second wheel cover **310** and a second intermediate member **320**. In addition, the mobile robot apparatus **1** may further include the first drive device **400**, the second drive device **500**, and a processor **600**.

(47) The first and second wheel covers **210** and **310** may be respectively connected to the first and second intermediate members **220** and **320**, which do not rotate relative to the body **100**, to be rotatable relative to body **100**. In addition, the body **100** may be configured such that opposite side surfaces are respectively connected to the first and second intermediate members **220** and **320** such that the intermediate members are vertically movable relative to body **100**.

(48) That is, the first and second wheel covers **210** and **310** may rotate relative to the first and second intermediate members **220** and **320**, and the body **100** may move vertically relative to the first and second intermediate members **220** and **320**.

(49) The first and second intermediate members **220** and **320** may be disposed between the wheel covers **210** and **310** and the body **100**, and may support the first drive device **400**, racks **530** and **560**, and guide rails **221** and **321**.

(50) The first drive device **400** may provide driving force to each of the first and second wheels **200** and **300**. The first drive device **400** may include a first motor **410** which provides driving force to the first wheel **200** and a second motor **420** which provides driving force to the second wheel **300**.

(51) The first motor **410** may be disposed on the first intermediate member **220** and rotate the first wheel cover **210**, and the second motor **420** may be disposed on the second intermediate member **320** and rotate the second wheel cover **310**.

(52) The second drive device may include racks **530** and **560**, which are disposed in at least one from among the first and second wheels **200** and **300**, and pinions **520** and **550**, which are disposed in the body **100** to engage with the racks **530** and **560**.

(53) Specifically, the second drive device **500** may include a third motor **510**, a first pinion **520**, and a first rack **530**. The first pinion **520** may be disposed in the body **100** to engage with the first rack **530**, and the first rack **530** may be disposed in the first wheel **200**.

(54) When the third motor **510** rotates the first pinion **520**, the first pinion **520** may move along the first rack **530**. Accordingly, the body **100** may move vertically relative to the first wheel **200**.

(55) In addition, the second drive device **500** may include a fourth motor **540**, a second pinion **550**, and a second rack **560**. The second pinion **550** may be disposed in the body **100** to engage with the second rack **560**, and the second rack **560** may be disposed in the second wheel **300**.

(56) When the fourth motor **540** rotates the second pinion **550**, the second pinion **550** may move along the second rack **560**. Accordingly, the body **100** may move vertically relative to the second

wheel **300**.

(57) The first and second racks **530** and **560** may be vertically disposed relative to surface **150**. Accordingly, the body **100** may move in a vertical direction relative to the surface **150** and to the center axis **X1** of the first and second wheels **200** and **300**.

(58) In addition, the first and second racks **530** and **560** may be respectively disposed at the first and second intermediate members **220** and **320**. Accordingly, because the first and second racks **530** and **560** maintain a certain shape regardless of the rotation of the first and second wheel covers **210** and **310**, the body **100** may stably move vertically relative to the first and second intermediate members **220** and **320**.

(59) As described above, as the second drive device includes two pairs of motors **510** and **540**, pinions **520** and **550**, and racks **530** and **560**, the body **100** may stably move vertically, and when the mobile robot apparatus **1** is turned, the body **100** may be tilted and an overturning of the body **100** may be prevented as is described below in reference to FIG. **12** and FIG. **13**.

(60) In addition, the mobile robot apparatus **1** may further include guide rails **221** and **321** which are disposed in at least one from among the first and second intermediate members **220** and **320**, and the body **100** may include blocks **110** and **120** which are moveable along the guide rails **221** and **321** at a side surface.

(61) Specifically, a first guide rail **221** may be disposed at the first intermediate member **220** and a second guide rail **321** may be disposed at the second intermediate member **320**. The first and second guide rails **221** and **321** may be vertically disposed relative to surface **150**.

(62) In addition, the body **100** may include a first block **110** which is moveable along the first guide rail **221** at one side surface and a second block **120** which is moveable along the second guide rail **321** at the other side surface.

(63) As the first and second blocks **110** and **120** of the body **100** are respectively joined to the first and second guide rails **221** and **321** and moved, the body **100** may move vertically relative to the first and second intermediate members **220** and **320**. In addition, because the first and second guide rails **221** and **321** guide a movement pathway of the body **100**, the body **100** may move vertically more stably.

(64) The mobile robot apparatus **1** may include the processor **600** for controlling an operation of the mobile robot apparatus **1**, and include a sensor **102** for recognizing the surrounding environment and a communication device for communicating with other electronic devices.

(65) The processor **600** may control the overall operation of the mobile robot apparatus **1**. To this end, the processor may include one or more from among a central processing unit (CPU), an application processor (AP), or a communication processor (CP). The processor may be a micro control unit (MCU).

(66) The processor **600** may operate an operating system or an application to control hardware or software elements connected to the processor **600**, and may perform various data processing and calculations. In addition, the processor **600** may load and process commands or data received from at least one from among other elements on a volatile memory, and store various data in a non-volatile memory.

(67) The processor **600** may receive surrounding environment information of the mobile robot apparatus **1** received from the camera **101** and at least one sensor **102**, and control the first drive device **400** and the second drive device **500** based on the received information. Descriptions of the processor **600** controlling the first and second drive devices **400** and **500** will be described in detail below.

(68) FIG. **10** is a front, cross-sectional view of the mobile robot apparatus **1** shown in FIG. **1** taken along line II-II. FIG. **11** is a front, cross-sectional view of the mobile robot apparatus **1** shown in FIG. **4** taken along line III-III.

(69) Referring to FIG. **10** and FIG. **11**, when the mobile robot apparatus **1** is required to travel at high speed, or is required to avoid obstacles, the processor **600** may control the second drive device

500 for the body **100** to move body **100** up relative to the center axis of the first and second wheels **200** and **300**. In addition, the processor **600** may control the second drive device **500** for the caster **103** to be selectively in contact with the surface **150** or to be spaced apart from the surface **150**.
(70) Specifically, the processor **600** may control the third and fourth motors **510** and **540** for the first pinion **520** to rotate in a first direction (counter-clockwise direction in FIG. **11**), and the second pinion **550** to rotate in a second direction (clockwise direction in FIG. **11**) which is opposite from the first direction.

(71) Accordingly, because the first and second pinions **520** and **550** respectively rotate to move upward along the first and second racks **530** and **560**, the body **100** may also move upward relative to the center axis of the first and second wheels **200** and **300**.

(72) FIG. **12** and FIG. **13** are views illustrating a body **100** being tilted when the mobile robot apparatus **1** is turned. Referring to FIG. **12** and FIG. **13**, the processor **600** may control the second drive device **500** for the body **100** to be tilted in a direction to which the mobile robot apparatus **1** is turned.

(73) For example, when the mobile robot apparatus **1** is turned to a front left side, the processor **600** may control the third and fourth motors **510** and **540** such that the first pinion **520** rotates in the first direction (counter-clockwise direction in FIG. **11**), and the second pinion **550** does not rotate. However, the control process of the processor **600** is not limited thereto, and the processor **600** may control the third and fourth motors **510** and **540** for the first and second pinions **520** and **550** to rotate in opposite directions from each other so as to tilt the body **100** at a greater angle.

(74) Accordingly, because the first pinion **520** moves to the upper side relative to the first rack **530**, the body **100** may be tilted in a direction to which the mobile robot apparatus **1** is turned ($-X$ direction in FIG. **13**). That is, because the body **100** is tilted in an opposite direction from a centrifugal force based on turning, the overturning of the mobile robot apparatus **1** may be prevented.

(75) FIGS. **14**, **15**, and **16** are views illustrating a process of the body **100** returning to its original position when having fallen.

(76) Referring to FIGS. **14** to **16**, based on the body **100** falling as it rotates in a first direction **R1** while the mobile robot apparatus **1** is moving in the two-point support mode described above, the body **100** may be in contact with the surface **150** from a first contact point **B**. At this time, the first and second wheels **200** and **300** may be in contact with the surface from a second contact point **W**.

(77) The processor **600** may control the second drive device **500** for the center axis **X1** of the first and second wheels **200** and **300** to move toward the center of body **100** until the center of mass **M** of the body **100** is positioned outside of a section **C** between the first contact point **B** of the body **100** and the second contact point **W** of the first and second wheels **200** and **300**.

(78) Specifically, the body **100** may be stopped in its place, and the center axis **X1** of the first and second wheels **200** and **300** may move in a direction toward the center of mass **M** of body **100**. Accordingly, as the second contact point **W** of the first and second wheels **200** and **300** becomes closer with the first contact point **B** of the body **100**, the center of mass **M** of the body **100** may be positioned outside of the section **C** between the first and second contact points **B** and **W**.

(79) The body **100** may be applied with a restoring torque in the second direction **R2** that is opposite from the first direction **R1** to which the body fell by gravity which acts on the center of mass **M** positioned at the outside of the contact section **C** based on the contact section **C**. Accordingly, the mobile robot apparatus **1** may again return to its position prior to falling as shown in FIG. **14**.

(80) The above-described control process is based on the scenario wherein body **100** falls in the first direction **R1** toward the “front” direction, but the direction to which the body **100** falls is not limited thereto, and may fall toward the rear direction. Even when the body **100** falls toward the rear direction, the above-described control process may be identically applied, and the mobile robot apparatus **1** may return to its position prior to falling.

(81) While example embodiments of the disclosure have been illustrated and described, it will be understood that the various example embodiments are intended to be illustrative, not limiting. It will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the true spirit and full scope of the disclosure, including the appended claims and their equivalents.

Claims

1. A mobile robot apparatus comprising: a body; a first wheel provided at a first side surface of the body; a second wheel provided at a second side surface of the body opposite to the first side surface; a first drive device configured to provide a driving force to each of the first wheel and the second wheel; a second drive device comprising a rack in at least one of the first wheel and the second wheel and a pinion in the body and engaged with the rack, wherein the second drive device is configured to move the body in a vertical direction relative to at least one of a first center axis of the first wheel and a second center axis of the second wheel by rotating the pinion; and at least one processor configured to, individually or collectively, control the second drive device to move the body to contact a surface on which the mobile robot apparatus is disposed, or move the body away from the surface on which the mobile robot apparatus is disposed.
2. The mobile robot apparatus of claim 1, wherein the first drive device comprises: a first motor configured to provide a first driving force to the first wheel, and a second motor configured to provide a second driving force to the second wheel.
3. The mobile robot apparatus of claim 1, wherein the rack is arranged vertically relative to the surface on which the mobile robot apparatus is disposed.
4. The mobile robot apparatus of claim 1, wherein the rack comprises: a first rack provided in the first wheel; and a second rack provided in the second wheel, and wherein the pinion comprises: a first pinion engaged with the first rack, and a second pinion engaged with the second rack.
5. The mobile robot apparatus of claim 4, wherein the second drive device further comprises: a third motor configured to rotate the first pinion, and a fourth motor configured to rotate the second pinion.
6. The mobile robot apparatus of claim 4, wherein the first wheel comprises: a rotatable first wheel cover; and a first intermediate member provided between the rotatable first wheel cover and the body, wherein the second wheel comprises: a rotatable second wheel cover; and a second intermediate member provided between the rotatable second wheel cover and the body, wherein the first rack is provided on the first intermediate member, and wherein the second rack is provided on the second intermediate member.
7. The mobile robot apparatus of claim 6, further comprising: a guide rail provided in at least one of the first intermediate member and the second intermediate member, wherein the body further comprises a block configured to be movable along the guide rail.
8. The mobile robot apparatus of claim 7, wherein the guide rail is oriented vertically relative to the surface on which the mobile robot apparatus is disposed.
9. The mobile robot apparatus of claim 1, further comprising: a camera provided at the body and configured to capture an image of a surrounding environment of the mobile robot apparatus.
10. The mobile robot apparatus of claim 1, further comprising: a caster provided at a bottom surface of the body and configured to be rotatable.
11. The mobile robot apparatus of claim 10, wherein a rotational axis of the caster is parallel with the first center axis of the first wheel or the second center axis the second wheel.
12. The mobile robot apparatus of claim 10, wherein the at least one processor is further configured to, individually or collectively, control the second drive device to selectively cause the caster to come into contact with or to be spaced apart from the surface on which the mobile robot apparatus is disposed.

13. The mobile robot apparatus of claim 1, wherein the at least one processor is further configured to, individually or collectively, control the second drive device to tilt the body in a direction to which the mobile robot apparatus is turning.

14. The mobile robot apparatus of claim 1, wherein the at least one processor is further configured to, individually or collectively, control the second drive device, based on the body contacting the surface on which the mobile robot apparatus is disposed, to move the first and the second wheels toward the body until a center of mass of the body is positioned outside a section between a contact point of the body with the surface on which the mobile robot apparatus is disposed and a contact point of the first and the second wheels with the surface on which the mobile robot apparatus is disposed.
